

BLM1612 - Circuit Theory

The Instructors:

Dr. Öğretim Üyesi Erkan Uslu

euslu@yildiz.edu.tr

Dr. Öğretim Üyesi Hamza Osman İlhan

hoilhan@yildiz.edu.tr

Lab Assistants:

Arş. Gör. Hasan Burak Avcı

<http://avesis.yildiz.edu.tr/hbavci/>

Arş. Gör. Kübra Adalı

<http://avesis.yildiz.edu.tr/adalik/>

Arş. Gör. Alper Egitmen

<http://avesis.yildiz.edu.tr/aegitmen/>

Units of Measurement
Basic Terminologies in Circuits
Circuit Components
Ohm's Law

Units of Measurement

- As engineers, we deal with measurable quantities.
- Measurement must be communicated in a standard language.
- The International System of Units (SI),
 - adopted by the General Conference on Weights and Measures in 1960.
- In SI, there are seven principal units from which the units of all other physical quantities can be derived.

Units of Measurement

- Six basic SI units and one derived unit relevant to this text.

Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd
Charge	coulomb	C

Units of Measurement

- One great advantage of the SI unit is that it uses prefixes based on the power of 10 to relate larger and smaller units to the basic unit.
- For example, the following are expressions of the same distance in meters (m):
 - 600 000 000 mm
 - 600 000 m
 - 600 km

The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

Units of Measurement

- The numerical value substituted into an equation must have the unit of measurement specified by the equation.
- For example,
 - consider the equation for the velocity $v = d / t$.
 - v : velocity, d : distance, t : time
 - Assume that the following data are obtained for a moving object: $d = 4000$ m, $t = 1$ min and v is desired in km per hour.
 - Incorrect answer:
 - $v = 4000 / 1 = 4000$ kmh
 - Correct answer:
 - $v = 4000 \times 10^{-3} / (1/60) = 240$ kmh

Units of Measurement

- Before substituting numerical values into an equation, be absolutely sure of the following:
 - Each quantity has the proper unit of measurement as defined by the equation.
 - The proper magnitude of each quantity as determined by the defining equation is substituted.
 - Each quantity is in the same system of units (or as defined by the equation).
 - The magnitude of the result is of a reasonable nature when compared to the level of the substituted quantities.
 - The proper unit of measurement is applied to the result.

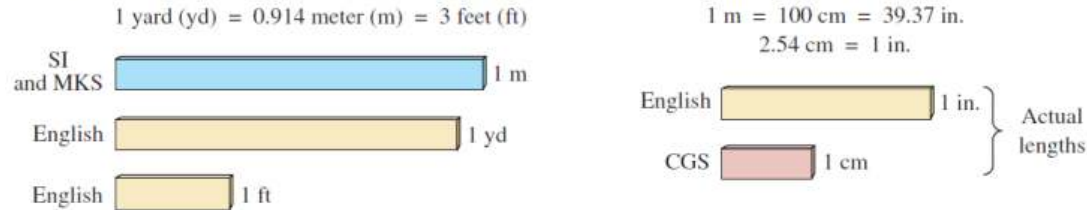
Systems of Units

- Comparison of the English and metric systems of units.

English	Metric		SI
	MKS	CGS	
<i>Length:</i> Yard (yd) (0.914 m)	Meter (m) (39.37 in.) (100 cm)	Centimeter (cm) (2.54 cm = 1 in.)	Meter (m)
<i>Mass:</i> Slug (14.6 kg)	Kilogram (kg) (1000 g)	Gram (g)	Kilogram (kg)
<i>Force:</i> Pound (lb) (4.45 N)	Newton (N) (100,000 dynes)	Dyne	Newton (N)
<i>Temperature:</i> Fahrenheit (°F) $\left(= \frac{9}{5}^{\circ}\text{C} + 32 \right)$	Celsius or Centigrade (°C) $\left(= \frac{5}{9} (^{\circ}\text{F} - 32) \right)$	Centigrade (°C)	Kelvin (K) $\text{K} = 273.15 + ^{\circ}\text{C}$
<i>Energy:</i> Foot-pound (ft-lb) (1.356 joules)	Newton-meter (N•m) or joule (J) (0.7376 ft-lb)	Dyne-centimeter or erg (1 joule = 10^7 ergs)	Joule (J)
<i>Time:</i> Second (s)	Second (s)	Second (s)	Second (s)

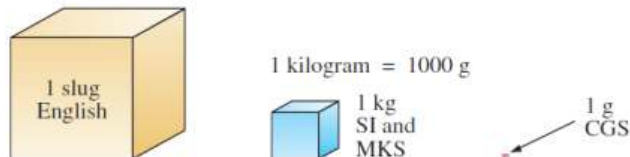
Comparison of units of the various systems of units

Length:

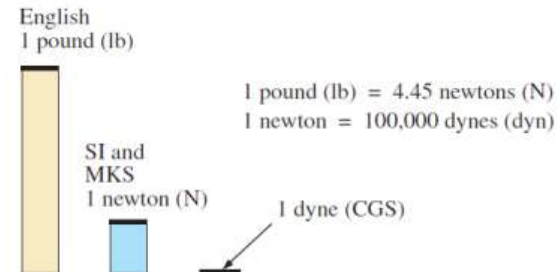


Mass:

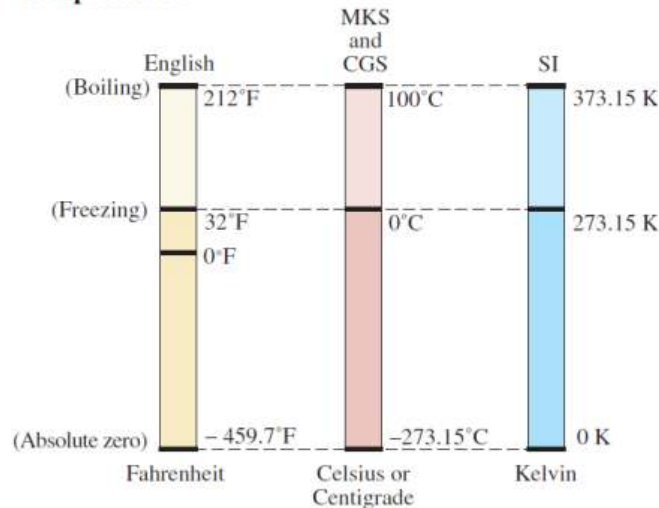
1 slug = 14.6 kilograms



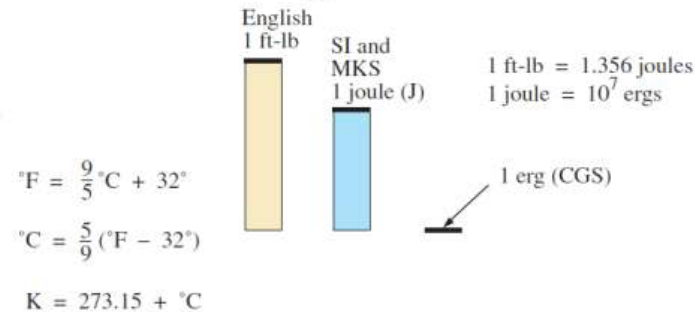
Force:



Temperature:



Energy:



Standards of some units

- The **meter** is defined with reference to the speed of light in a vacuum, which is **299792458** m/s.
 - It was originally defined in 1790 to be 1/**10000000** the distance between the equator and either pole at sea level, a length preserved on a platinum-iridium bar at the International Bureau of Weights and Measures at Sèvres, France.
- The **kilogram** is defined as a mass equal to 1000 times the mass of one cubic centimeter of pure water at 4°C.
 - This standard is preserved in the form of a platinum-iridium cylinder in Sèvres.

Standards of some units

- The **second** is redefined in 1967 as 9192631770 periods of the electromagnetic radiation emitted by a particular transition of cesium atom.
 - It was originally defined as $1/86400$ of the mean solar day.
 - However, Earth's rotation is slowing down by almost 1 second every 10 years.

Significant Figures, Accuracy, Round off

- Two types of numbers:
 - **Exact**
 - For example 12 apples
 - **Approximate**
 - Any reading obtained in the laboratory should be considered approximate
- The **precision** of a reading can be determined by the number of **significant figures** (**digits**) present.
- **Accuracy** refers to the closeness of a measured value to a standard or known value
- For approximate numbers, there is often a need to **round off** the result
 - that is, you must decide on the appropriate level of accuracy and alter the result accordingly.
 - For example, $3.186 \cong 3.19 \cong 3.2$

Powers of ten

- To express very large and very small numbers
- The notation used to represent numbers that are integer powers of ten is as follows:

$$1 = 10^0 \qquad 1/10 = 0.1 = 10^{-1}$$

$$10 = 10^1 \qquad 1/100 = 0.01 = 10^{-2}$$

$$100 = 10^2 \qquad 1/1000 = 0.001 = 10^{-3}$$

$$1000 = 10^3 \qquad 1/10,000 = 0.0001 = 10^{-4}$$

Powers of ten

- Some important mathematical equations and relationships pertaining to powers of ten:

$$\frac{1}{10^n} = 10^{-n} \quad \frac{1}{10^{-n}} = 10^n$$

$$(10^n)(10^m) = (10)^{(n+m)}$$

$$\frac{10^n}{10^m} = 10^{(n-m)}$$

$$(10^n)^m = 10^{nm}$$

Powers of ten

- Addition and subtraction

$$A \times 10^n \pm B \times 10^n = (A \pm B) \times 10^n$$

- Multiplication

$$(A \times 10^n)(B \times 10^m) = (A)(B) \times 10^{n+m}$$

- Division

$$\frac{A \times 10^n}{B \times 10^m} = \frac{A}{B} \times 10^{n-m}$$

- Power

$$(A \times 10^n)^m = A^m \times 10^{nm}$$

Scientific notation vs. Engineering notation

- Scientific notation and engineering notation make use of powers of ten, with restrictions on the mantissa (multiplier) or scale factor (power of ten).
 - Scientific notation requires that the decimal point appear directly after the first digit greater than or equal to 1 but less than 10.
 - Engineering notation specifies that all powers of ten must be multiples of 3, and the mantissa must be greater than or equal to 1 but less than 1000.

Scientific notation vs. Engineering notation

- Scientific notation example:

$$\frac{1}{3} = 3.333333333333\text{E}-1 \quad \frac{1}{16} = 6.25\text{E}-2 \quad \frac{2300}{2} = 1.15\text{E}3$$

$$\frac{1}{3} = 3.33\text{E}-1 \quad \frac{1}{16} = 6.25\text{E}-2 \quad \frac{2300}{2} = 1.15\text{E}3$$

- Engineering notation example:

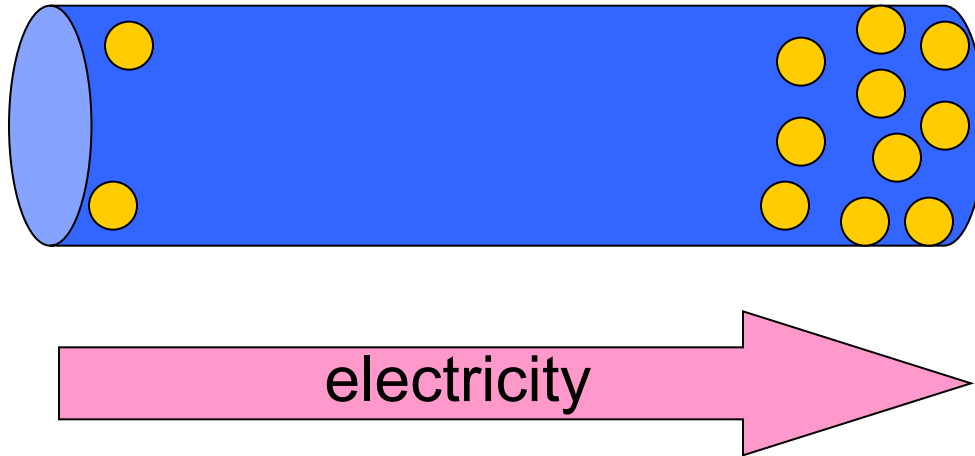
$$\frac{1}{3} = 333.3333333333\text{E}-3 \quad \frac{1}{16} = 62.5\text{E}-3 \quad \frac{2300}{2} = 1.15\text{E}3$$

$$\frac{1}{3} = 333.33\text{E}-3 \quad \frac{1}{16} = 62.50\text{E}-3 \quad \frac{2300}{2} = 1.15\text{E}3$$

Electricity

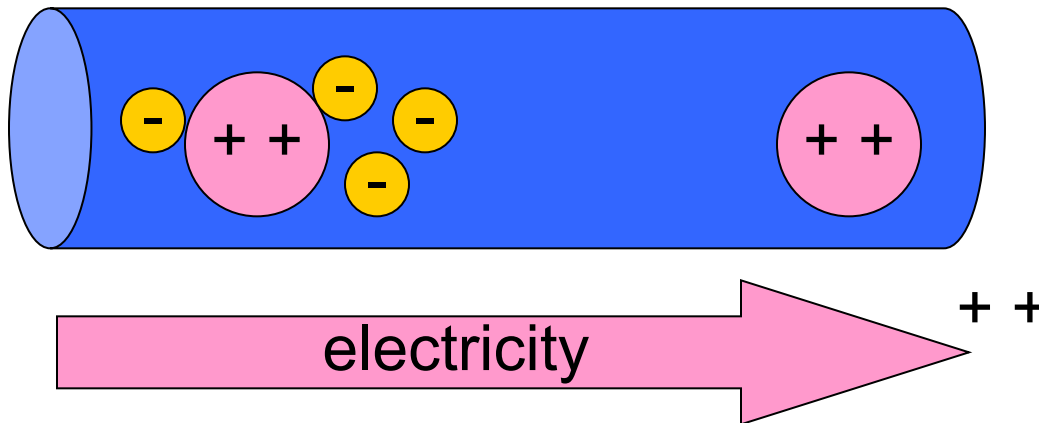
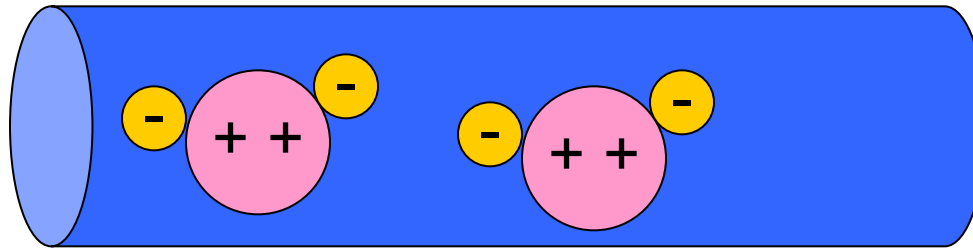
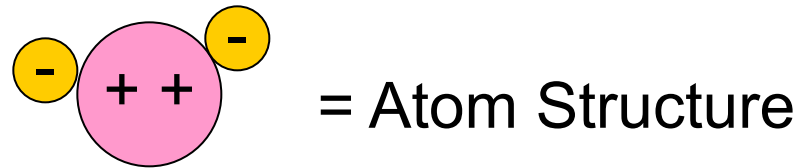
- Electricity is a result from the flow of electrons.

● = electron



- Electricity flows in the opposite direction of electron flow.

Electric Current vs. Electron Current



Electric current

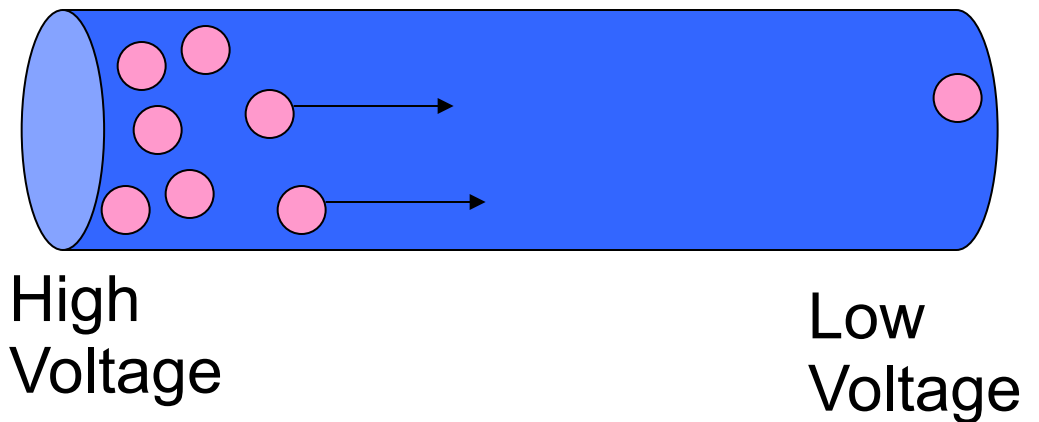
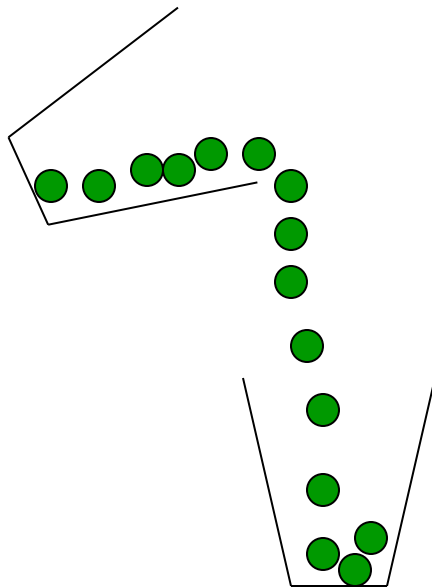
- We cannot see electric current.
- We need a metaphor.
- Which thing has similar property with electricity??



Water

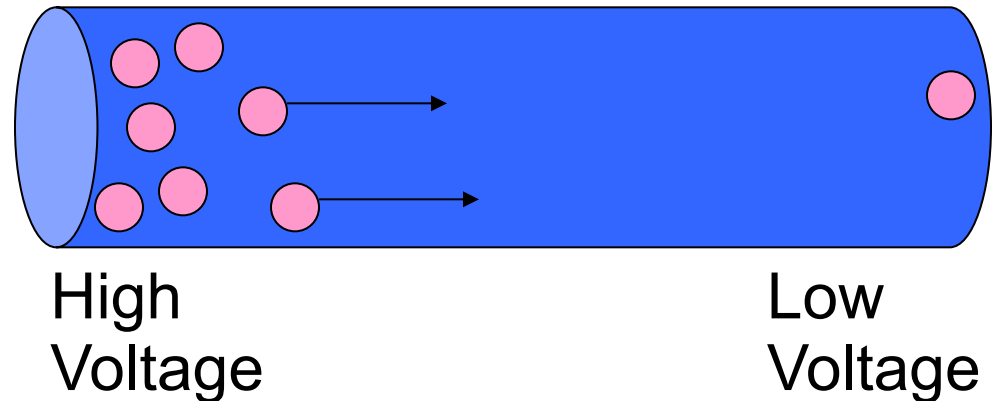
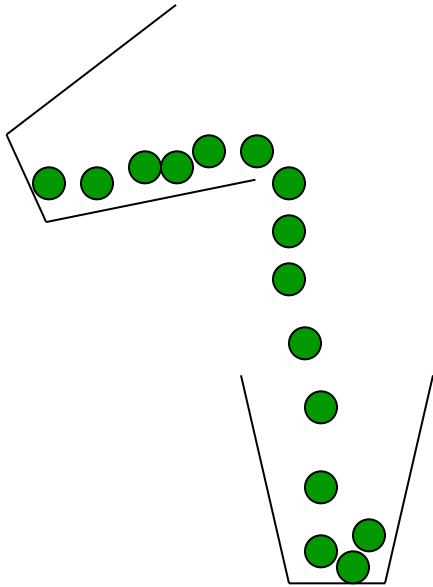
Electric current

- Electricity is similar to water flow.
 - Water flows from high level to low level.
 - Electricity flows from high voltage to low voltage.



Measurement of Electricity

- Since we use electricity to do work for us, how can we measure its energy?
- How can we measure the water power?
 - Think about a water gun.



- strong (fast, high kinetic energy)
- amount of water

Voltage
Current

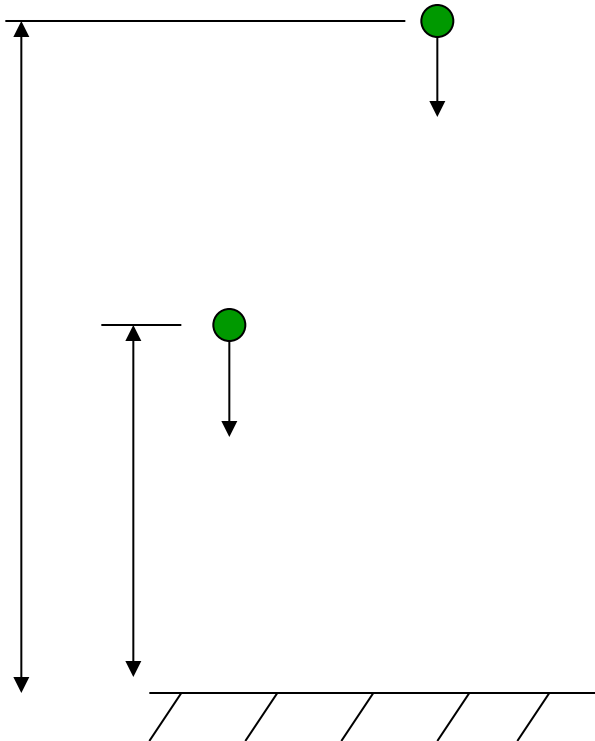
Measurement of Electricity

- Imagine the water power at the outlet

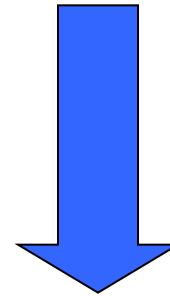


Electric Potential

- Which water drop has more impact force at the ground?



- Potential Energy-Height



transform

- Kinetic Energy-Velocity

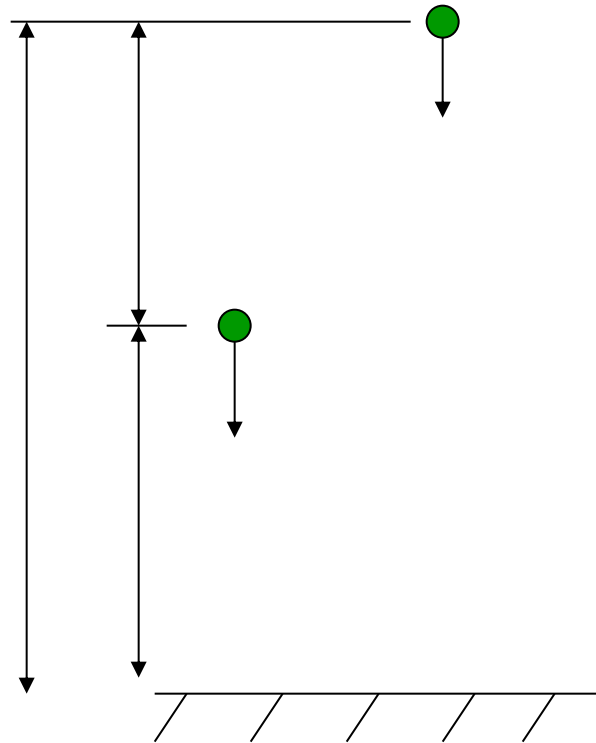
- Electric potential can be compared with the height of the water drop from the reference ground

Ground: Reference Point

- Normally, we measure height compared to the sea level.
- Also, electric potential at a point can be measured compared to the electric potential at the **ground**.
- Electric potential, or voltage has a unit **volt(V)**.
- Ground always has **0** volts.

Voltage

- Voltage is a difference of electric potential between 2 points

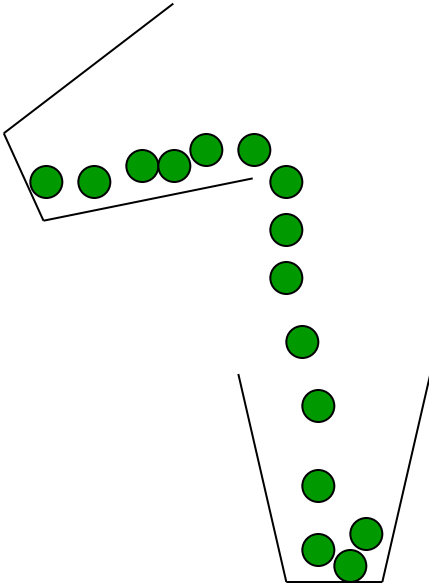


Unit: Volt

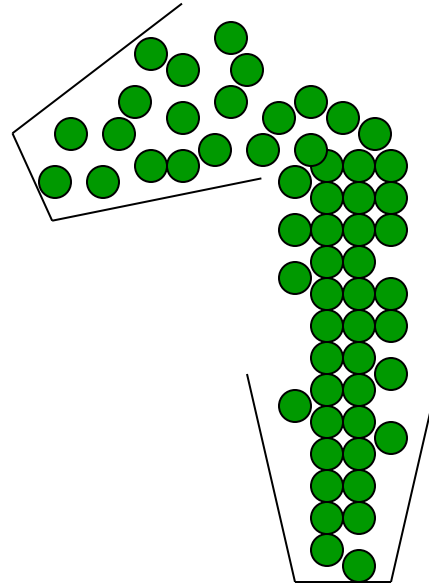
- Compare to the height of 2 water drops

Electric Current

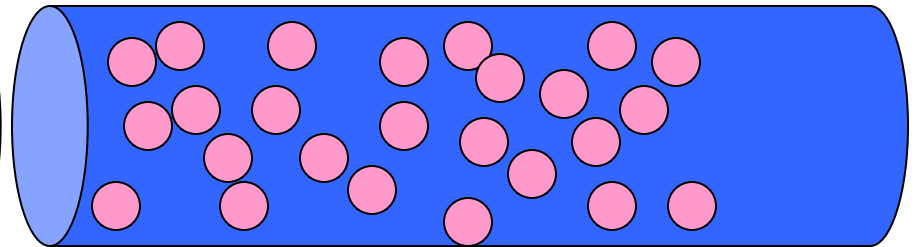
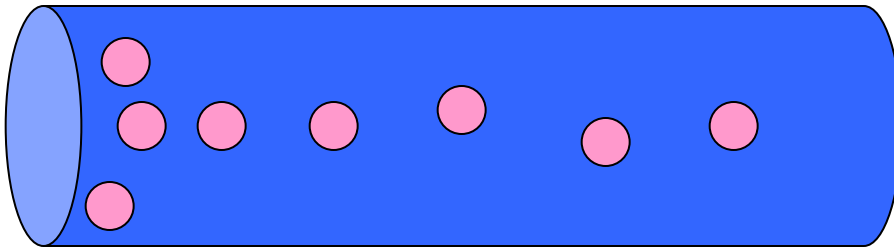
Unit: Ampere,
Amp (A)



Low current



High current

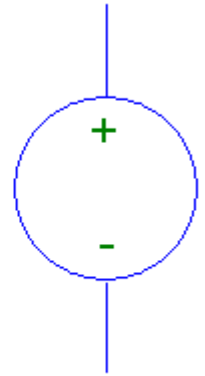
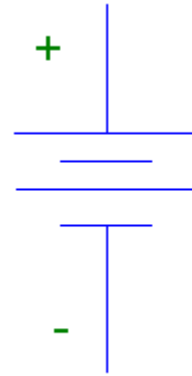


Circuit Components

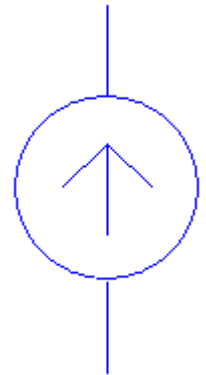
- Active elements
 - Independent power sources
 - voltage, current
 - Dependent power sources
 - voltage, current
- Passive Elements
 - Resistors
 - Capacitors
 - Inductors
- Measurement Devices
 - Ampermeters:
 - measure current
 - Voltmeters:
 - measure voltage
- Ground
 - reference point
- Electric Wire
- Switches
- Protective devices
 - Fuse

Independent Power Sources

- Independent voltage source outputs a **voltage**, either **dc** or **time varying**, to the circuit no matter how much current is required.



- Independent current source outputs a **dc** or **ac** current to the circuit no matter how much voltage is required.

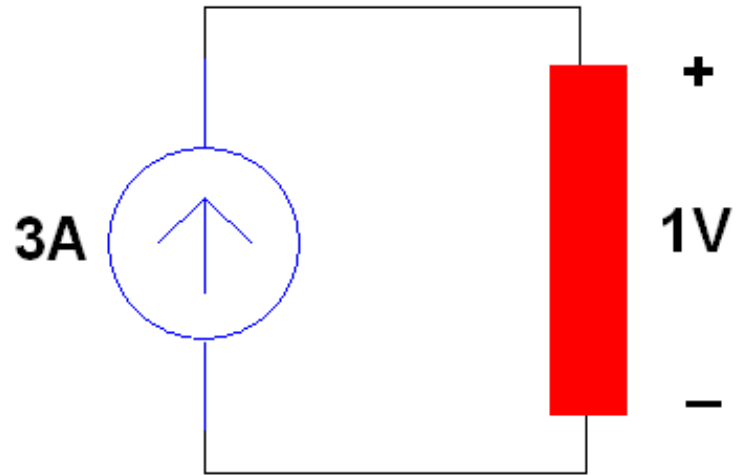


Independent Power Sources

- Current can flow in and out of an independent voltage source, but the polarity of the voltage is determined by the voltage source.
- There is always a voltage drop across the independent current source and the direction of positive current is determined by the current source.

Example 1

- 1V is dropped across some element (in red) and the wires to that element are connected directly to the independent current source.

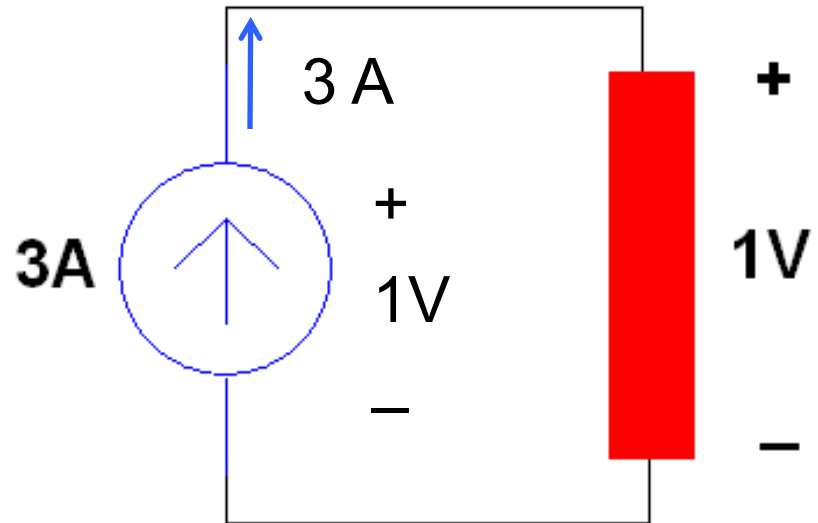


Example 1

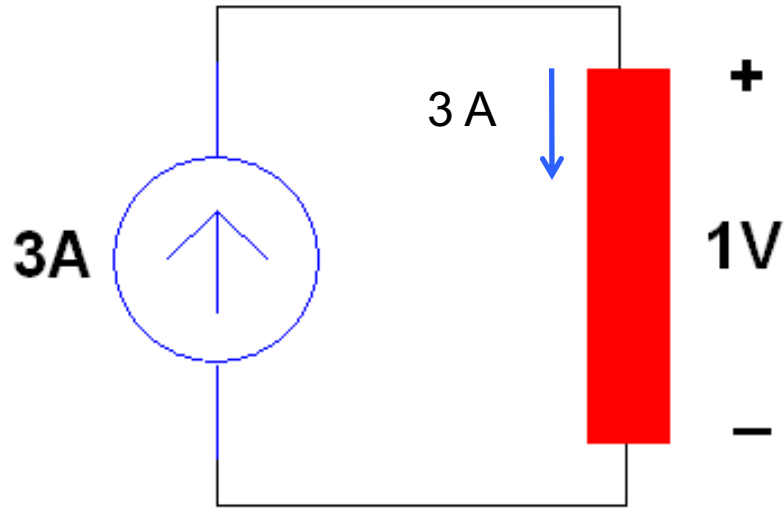
- This means that 1 V is also dropped across the independent current source. Therefore, the current source is generated $1 \text{ V}(3 \text{ A}) = 3 \text{ W}$ of power.

- Passive sign convention: When current leaves the + side of a voltage drop across the independent current source, the power associated with the current source is:

$$p = -3 \text{ A}(1 \text{ V}) = -3 \text{ W}$$



Example 1



- Conservation of energy means that the other element in red must be dissipating 3 W of power.

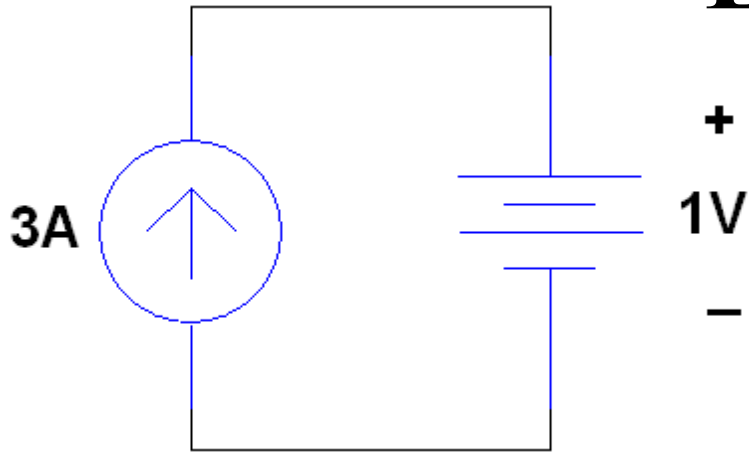
$$\sum p = p_{\text{current source}} + p_{\text{red element}} = 0$$

$$p_{\text{current source}} = -3 \text{ W}; \text{ therefore, } p_{\text{red element}} = 3 \text{ W}$$

- Passive sign convention:
When current enters the + side of a voltage drop across the element in red, the power associated with this element is:

$$p = 3 \text{ A}(1 \text{ V}) = 3 \text{ W}$$

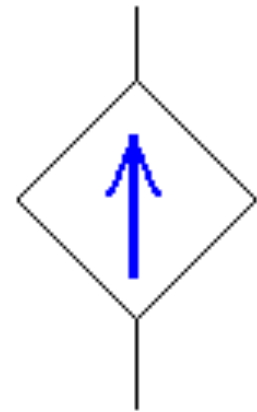
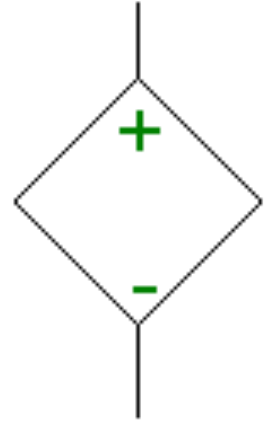
Example 1



- Suppose the red element was an independent voltage source.
- This means that the independent current source happens to be supplying power to the independent voltage source, which is dissipating power.
- This happens when you are charging a battery, which is considered to be an independent voltage source.

Dependent Power Sources

- Voltage controlled voltage source
 - (VCVS)
- Current controlled voltage source
 - (CCVS)
- Voltage controlled current source
 - (VCCS)
- Current controlled current source
 - (CCCS)



Passive Elements

- The magnitude of the voltage drop and current flowing through passive devices depends on the voltage and current sources that are present and/or recently attached to the circuit.
 - These components can dissipate power immediately or store power temporarily and later release the stored power back into the circuit.

Passive Components

Component	Symbol	Basic Measure (Unit)
Resistor 		Ohm (Ω)
Inductor 		Henry (H)
Capacitor 		Farad (F)

Other Basic Circuit Elements

- Electric wire



Symbol



- Ground



earth



chassis



ground



ground

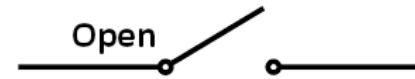


analog
ground



digital
ground

- Switch



- Fuse



Switches

- Switches are used to control whether a complete path is formed from an end of at least one power supply to the other end of the same power supply (closed circuit).
 - Current will only flow when there is a closed circuit.
- Switches can be mechanical, as are used on light switches in your home, or are electronic switches, which are semiconductor based.
- Electronic switches are used in TV sets, for example, to turn on the TV when an infrared optical signal from the remote control is detected.

Protective Devices

- Circuits that have carry dangerous levels of current and voltages are required to include fuses, circuit breakers, or ground fault detectors by federal and state electrical safety codes.
 - These protective devices are designed to create an open circuit, or a break in the round trip path in the circuit, when a malfunction of a component or other abnormal condition occurs.
 - The speed of response of the protective device, fast-acting or time-delay (slow-blow) is determine by the engineer, based upon the expected type of malfunction.

Wires

- Wires are assumed to have zero resistance; i.e., they are ideal conductors or short circuits.
 - The current carrying capability of a wire is determined by its diameter or cross-sectional area.
 - AWG, American wire gauge, is the standard followed in the US and is used to rate how much current a wire can safely carry.
 - The larger the gauge wire, the smaller its current carrying capability is.
 - The AWG standard includes copper, aluminum and other wire materials.
 - Typical household copper wiring is AWG number 12 or 14.
 - Telephone wire is usually 22, 24, or 26.
 - The higher the gauge number, the smaller the diameter and the thinner the wire.

AWG to square mm cross sectional area

American Wire Gauge (#AWG)	Diameter (inches)	Diameter (mm)	Cross Sectional Area (mm ²)
0000 (4/0)	0.460	11.7	107
000 (3/0)	0.410	10.4	85.0
00 (2/0)	0.365	9.27	67.4
0 (1/0)	0.325	8.25	53.5
1	0.289	7.35	42.4
2	0.258	6.54	33.6
3	0.229	5.83	26.7
4	0.204	5.19	21.1
5	0.182	4.62	16.8
6	0.162	4.11	13.3
7	0.144	3.67	10.6
8	0.129	3.26	8.36
9	0.114	2.91	6.63
10	0.102	2.59	5.26

American Wire Gauge (#AWG)	Diameter (inches)	Diameter (mm)	Cross Sectional Area (mm ²)
11	0.0907	2.30	4.17
12	0.0808	2.05	3.31
13	0.0720	1.83	2.63
14	0.0641	1.63	2.08
15	0.0571	1.45	1.65
16	0.0508	1.29	1.31
17	0.0453	1.15	1.04
18	0.0403	1.02	0.82
19	0.0359	0.91	0.65
20	0.0320	0.81	0.52
21	0.0285	0.72	0.41
22	0.0254	0.65	0.33
23	0.0226	0.57	0.26
24	0.0201	0.51	0.20
25	0.0179	0.45	0.16
26	0.0159	0.40	0.13

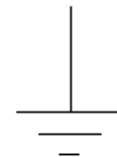
AWG to ohm/meter

Approximate resistance of copper wire^{[6]:27}

AWG	mΩ/ft	mΩ/m	AWG	mΩ/ft	mΩ/m	AWG	mΩ/ft	mΩ/m	AWG	mΩ/ft	mΩ/m
0	0.1	0.32	10	1	3.2	20	10	32	30	100	320
1	0.125	0.4	11	1.25	4	21	12.5	40	31	125	400
2	0.16	0.5	12	1.6	5	22	16	50	32	160	500
3	0.2	0.64	13	2	6.4	23	20	64	33	200	640
4	0.25	0.8	14	2.5	8	24	25	80	34	250	800
5	0.32	1	15	3.2	10	25	32	100	35	320	1000
6	0.4	1.25	16	4	12.5	26	40	125	36	400	1250
7	0.5	1.6	17	5	16	27	50	160	37	500	1600
8	0.64	2	18	6.4	20	28	64	200	38	640	2000
9	0.8	2.5	19	8	25	29	80	250	39	800	2500

Ground

- Earth ground is a ground that is physically connected to the earth, itself.
 - All homes have an earth ground
 - a wire connected to a metal pipe that is driven into the ground immediately next to the house.
 - Wires that have a green jacket or are bare copper are connected to this pipe.
- Reference ground or common is used in a circuit to indicate a point where the voltage in the circuit is equal to zero.



General Rules

- All points on a same electric wire have the same voltage.
- A voltage source always have voltage difference of its pins equal to its value.
- A current source always have current pass through it equal to its value.
- Ground always has zero voltage. (0 volts)

Electric Flow Rule

- Electric current flows from high voltage to low voltage when there is a path.
- Electric current can freely pass through electric wire.
- Electric current can flow through a resistor with the amount according to Ohm's law.
- Electric current can flow through a voltage source with the amount depended on other components in the circuit.
- Electric current can flow pass a current source according to its value.

Charge

- Electrical property of atomic particles
 - Electrons are negatively charged
 - Protons are positivity charged
- The absolute value of the charge on an electron is 1.6×10^{-19} C
- The symbol used is Q or q
 - Uppercase is used to denote a steady-state or constant value
 - Lowercase is used to denote an instantaneous value or time-varying quantity

Current

- The flow of charge through a cross-sectional area as a function of time or the time rate of change of charge
- Symbol used is I or i

$$i = \frac{dq}{dt}$$

$$Q = \int_{t_1}^{t_2} i \, dt$$

DC vs. AC

- DC (or dc) is the acronym for direct current.
 - The current remains constant with time.
 - Uppercase variables are used when calculating dc values.
- AC (or ac) is the acronym for alternating current.
 - Specifically, AC current varies sinusoidally with time and the average value of the current over one period of the sinusoid is zero.
 - Lowercase variables are used when calculating ac values.
 - Other time-varying currents exist, but there isn't an acronym defined for them.

Voltage (Potential Difference)

- The **electromotive force (emf)** that causes charge to move.
- 1 Volt = 1 Joule/1 Coulomb

$$V = \frac{dw}{dq}$$

Power

- The change in energy as a function of time is power, which is measured in **watts (W)**.

$$p = \frac{dw}{dt} = \left[\frac{dw}{dq} \right] \left[\frac{dq}{dt} \right] = vi$$

Energy

- Energy is the capacity to do work.

$$w = \int_{t_1}^{t_2} p \, dt = \int_{t_1}^{t_2} v \, i \, dt$$

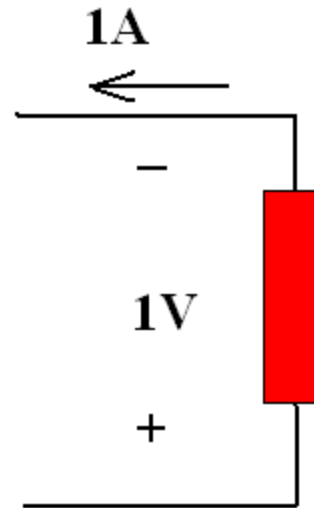
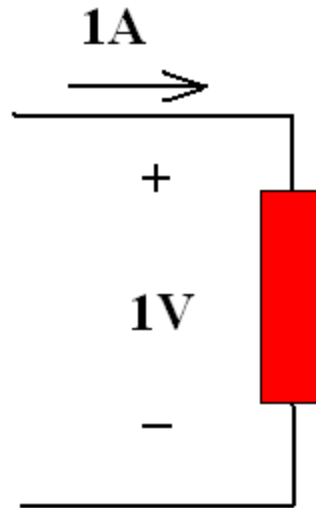
- Units for energy are kW-hr, which is what the electric company measures on your electric meter.

$$1 \text{ kW-hr} = 3.6 \text{ MJ.}$$

Positive vs. Negative Power

- Power consumed/dissipated by a component is positive power

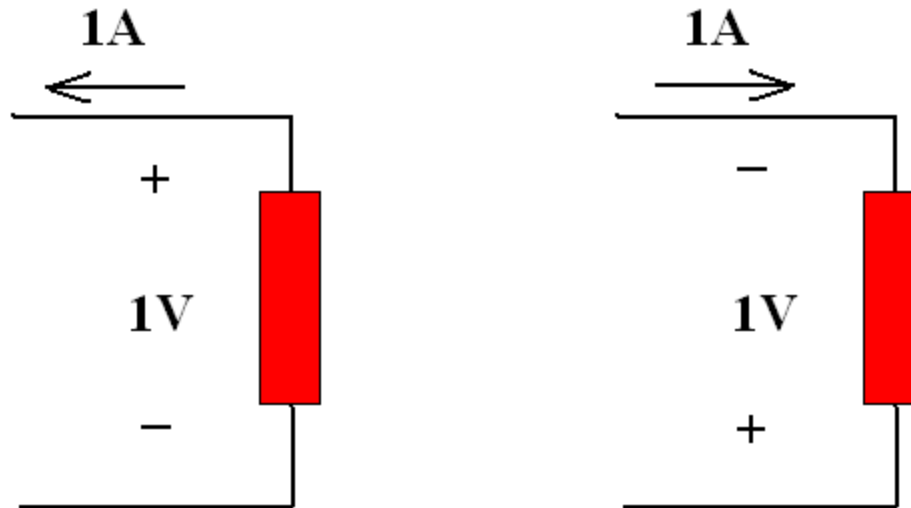
$$P = +1\text{W}$$



Positive vs. Negative Power

- Generated power has a negative sign

$$P = -1\text{ W}$$



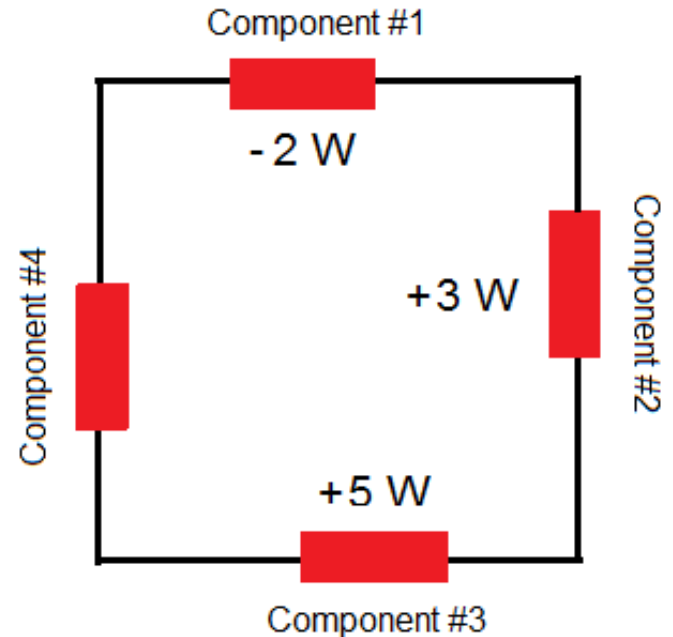
Conservation of Energy

- All power instantaneously consumed by components must be instantly generated by other components within the circuit.

$$\sum p = 0$$

Example

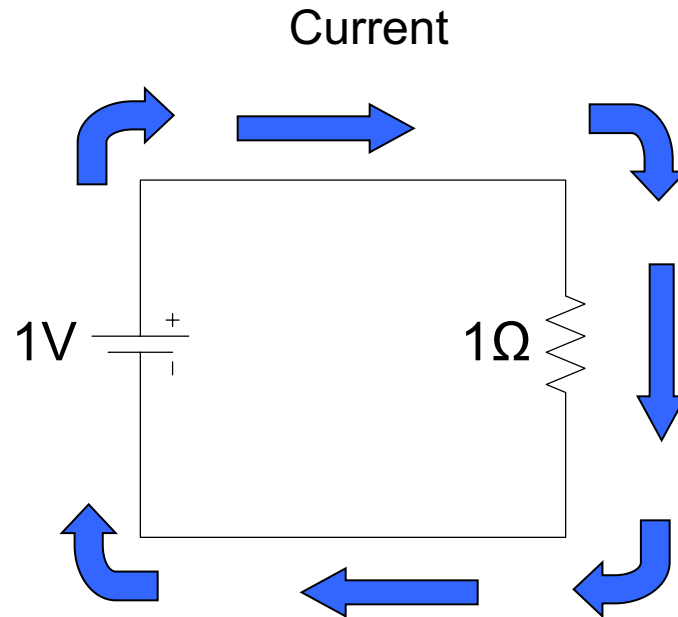
- There are 4 electrical components in the circuit shown to the right.
- Component #1 is generating 2 W of power and supplying this power to the circuit.
- Components #2 and #3 are consuming power.
- Component #2 is dissipating 3 W of power while Component #3 is dissipating 5 W of power.
- Component #4 must be generating 6 W of power in order to maintain the Conservation of Energy.



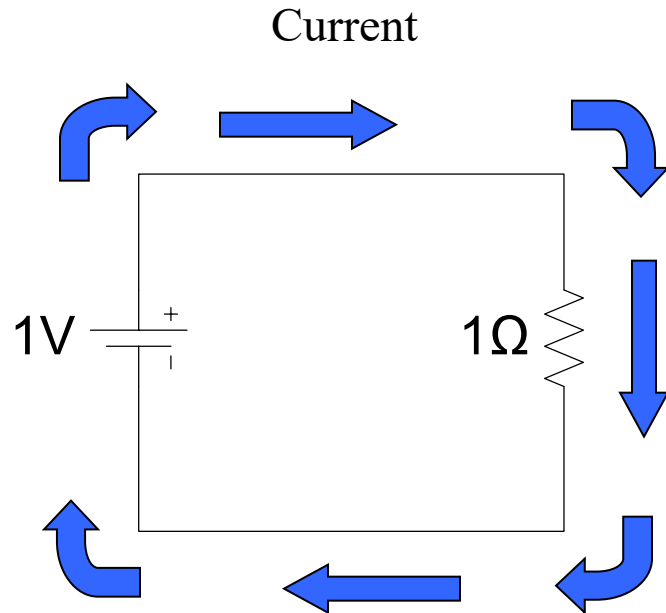
$$\sum p = p_{\text{Component\#1}} + p_{\text{Component\#2}} + p_{\text{Component\#3}} + p_{\text{Component\#4}} = 0$$

$$p_{\text{Component\#4}} = -(p_{\text{Component\#1}} + p_{\text{Component\#2}} + p_{\text{Component\#3}}) = -(-2 \text{ W} + 3 \text{ W} + 5 \text{ W}) = -6 \text{ W}$$

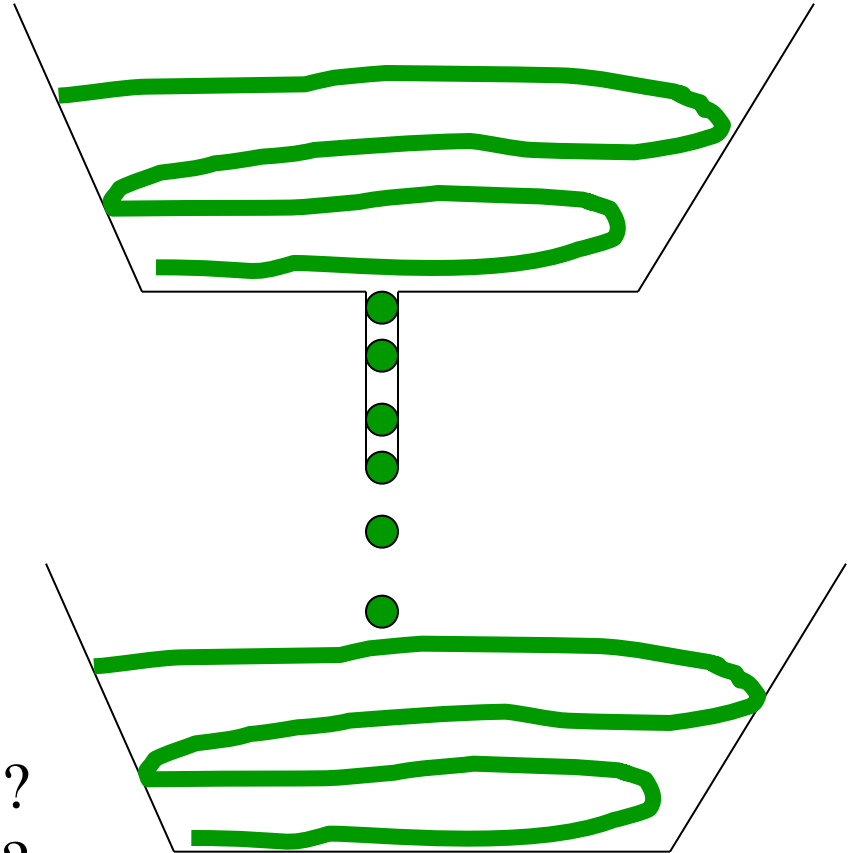
Simple DC Circuit



Metaphor



- Increasing V is compared to ?
- Increasing R is compared to ?

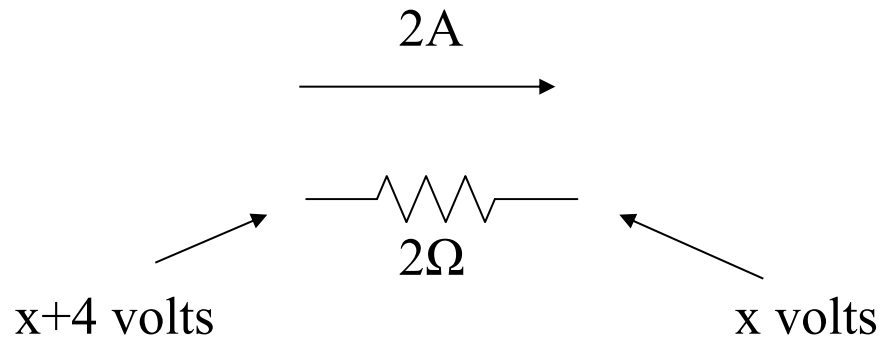


Ohm's Law

$$V = IR$$

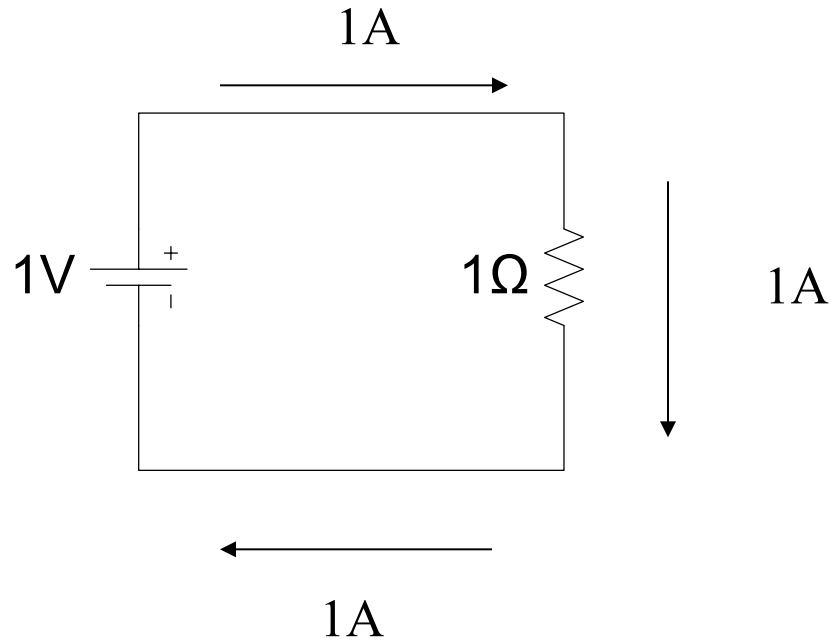
for using with a resistor only

Voltage (Volts) =
current (Amperes) x resistance (Ohms)



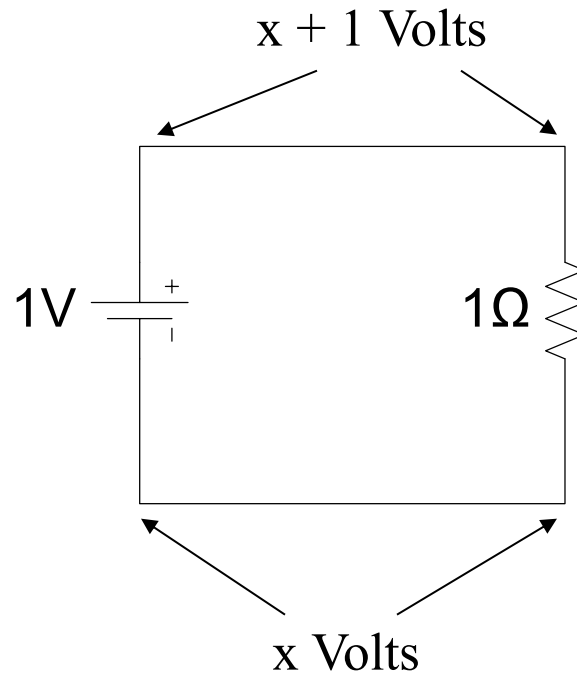
Note: (Theoretically) Electric wire has a resistance of 0 ohms

Electric Current



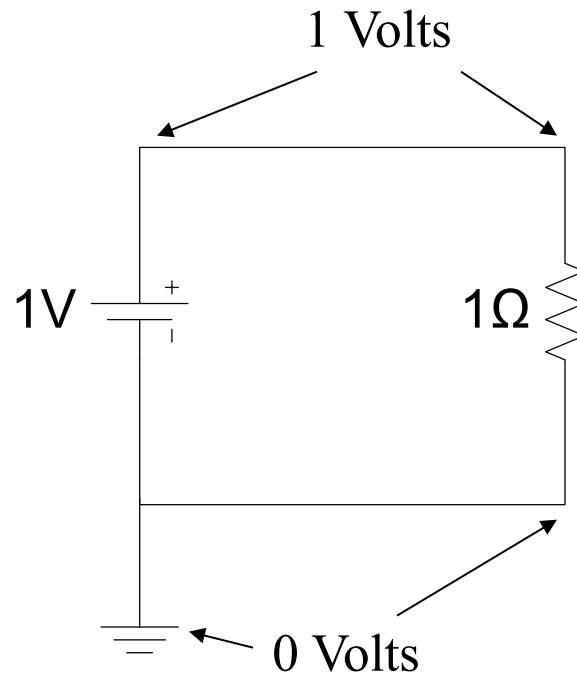
Every point in the circuit has current = $1A$

Electric Voltage



Ground

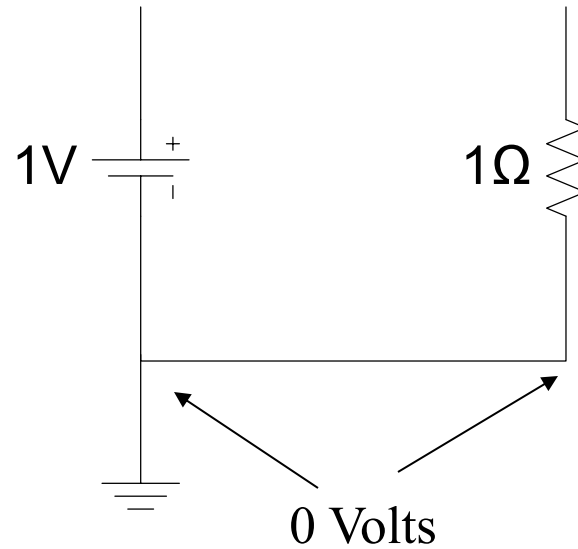
Ground = reference point always have voltage = 0 volts



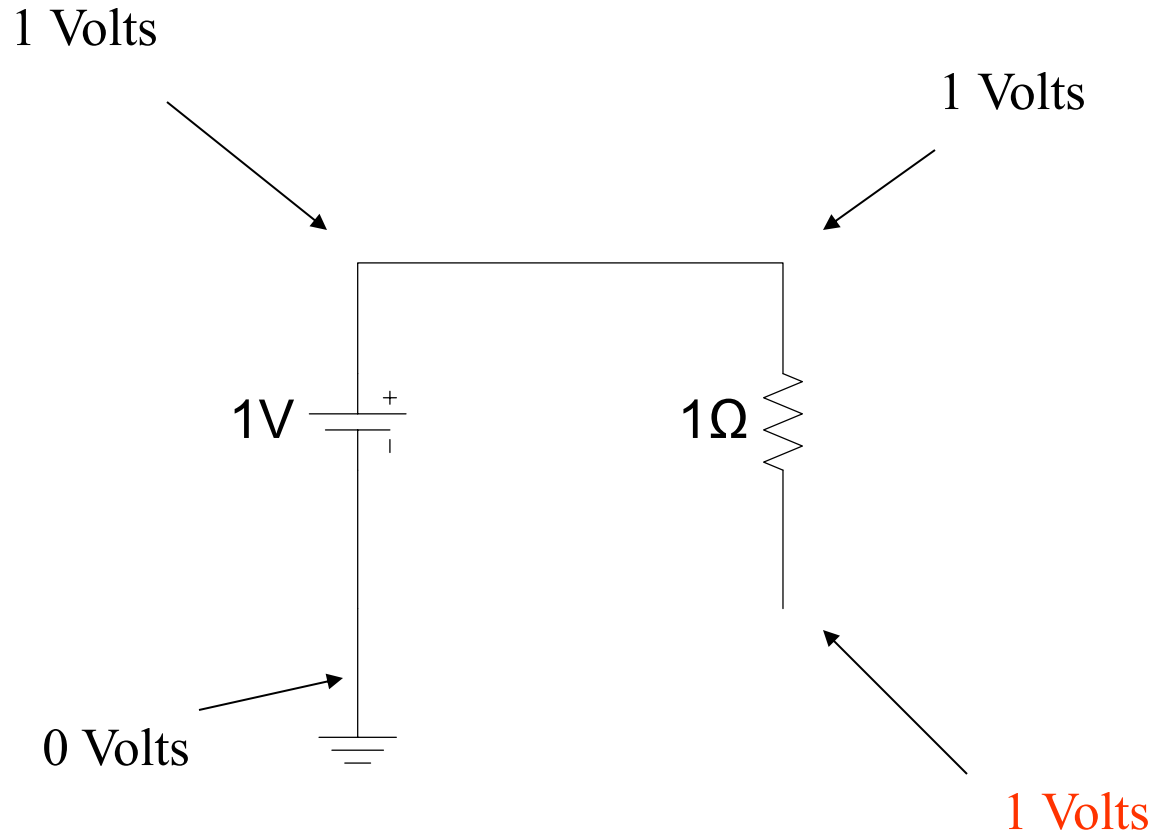
Electric Voltage (2)

1 Volts

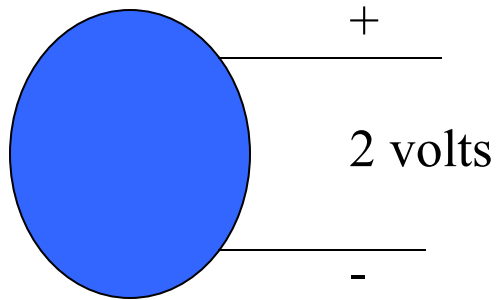
0 Volts



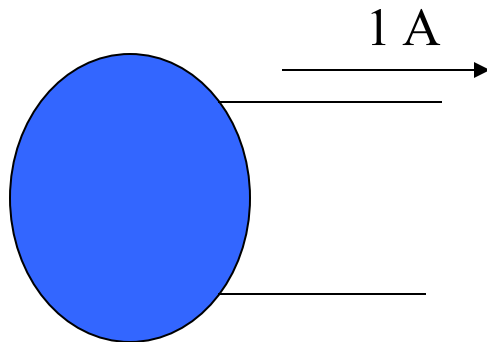
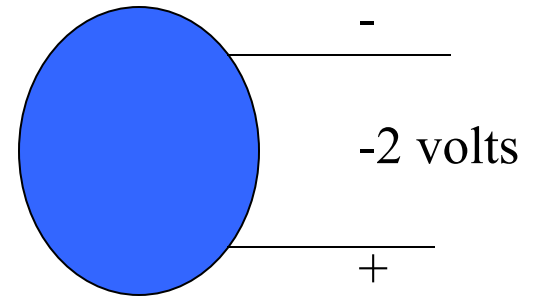
Electric Voltage (3)



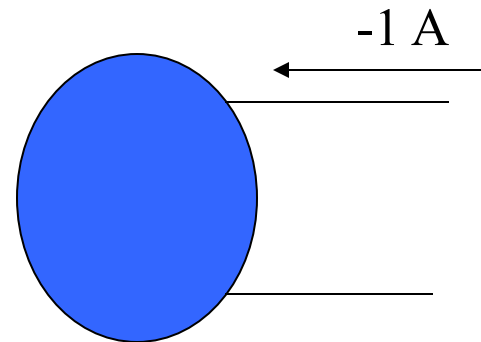
Negative Voltage and Current



Same as



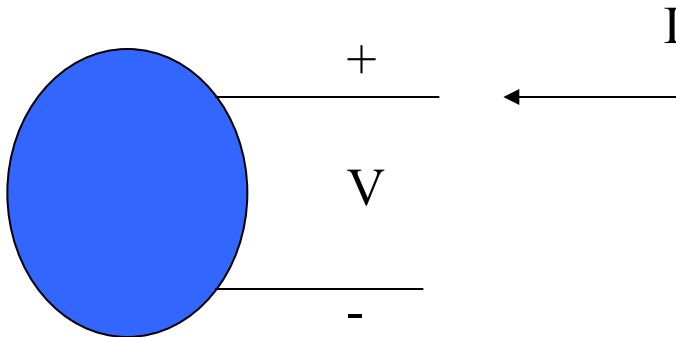
Same as



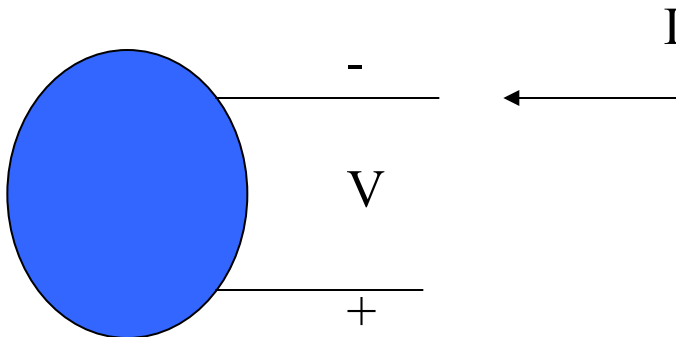
Power

Symbol P has a unit of Watt

$$P = VI$$



Absorb power



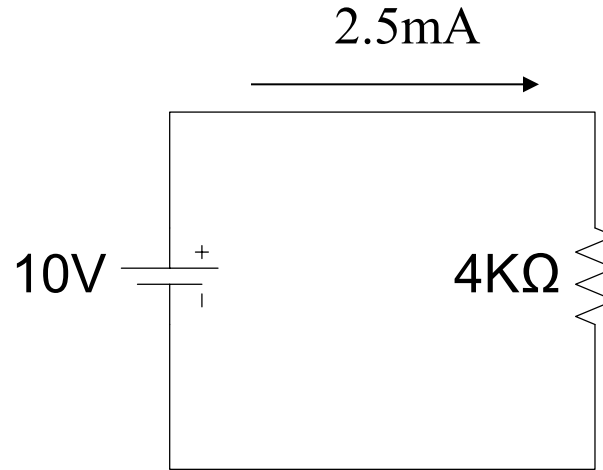
Generate power

Passive Sign Convention

Absorb power: Power has a sign +

Generate power: Power has a sign -

Example

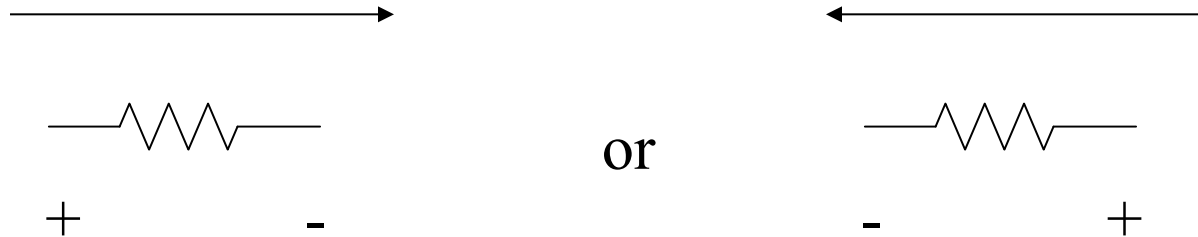


DC source generates power = $10\text{V} * -2.5\text{mA} = -25\text{mW}$

Resistor absorbs power = $10\text{V} * 2.5\text{mA} = 25\text{mW}$

Note: Resistors always absorb power but DC source can either generate or absorb power

Direction of Voltage & Current on Resistors



- Resistor always absorb power.
- Therefore, it always have current flow through it from high voltage pin to low voltage pin.